

Cauchy Integral Formula

COMPLEX ANALYSIS

BSC ENGINEERING MATHEMATICS

One of the most elegant and powerful results in complex analysis, the Cauchy Integral Formula reveals a remarkable property of analytic functions: the value of a function at any interior point is completely determined by its values on the surrounding boundary. This single theorem bridges geometry, calculus, and algebra in a way that is both mathematically profound and practically indispensable for engineers and applied mathematicians.



Prerequisites

Before engaging with the Cauchy Integral Formula, a solid understanding of several foundational concepts in complex analysis is essential. These ideas form the logical scaffolding upon which the theorem is built.

Analytic Functions

A function $f(z)$ is said to be *analytic* (or *holomorphic*) at a point z_0 if it is differentiable in some neighborhood of z_0 . Analyticity is a far stronger condition than real differentiability — it implies infinite differentiability and the existence of a convergent power series (Taylor series) in a neighborhood of every point in the domain. A function analytic throughout a region is called *entire* if that region is the entire complex plane.

Closed Contours

A **contour** is a piecewise-smooth directed curve in the complex plane. A *closed contour* is one that begins and ends at the same point. The standard notation is C , and we typically require C to be a *simple* closed curve — one that does not intersect itself. The interior of such a curve is well-defined by the Jordan Curve Theorem. Common examples include circles $|z - a| = r$ and rectangles traversed counterclockwise.

Cauchy–Goursat Theorem

The **Cauchy–Goursat Theorem** states that if f is analytic inside and on a simple closed contour C , then $\oint_C f(z) dz = 0$. This is the cornerstone of contour integration theory. The Cauchy Integral Formula can be viewed as a refinement of this result — it tells us what happens when the integrand has a simple pole at an interior point. Goursat's contribution was to remove the requirement of continuous derivative, proving the theorem using only the existence of f' .

Statement of the Cauchy Integral Formula

The Formula

Let f be analytic inside and on a simple closed positively oriented contour C . Then for any point a lying inside C :

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a} dz$$

Equivalently, solving for the integral:

$$\oint_C \frac{f(z)}{z - a} dz = 2\pi i f(a)$$

What This Means

The formula expresses the value of an analytic function at an *interior* point a as a contour integral taken over the *boundary* C .

This is a **representation formula** — it does not merely compute an integral; it establishes a deep structural relationship between a function's interior values and its boundary values.

Conditions for Validity

The Cauchy Integral Formula is not universally applicable – it requires three precise conditions to hold simultaneously. Violating any one of them invalidates the formula or requires modification (e.g., using the Residue Theorem for multiple singularities).



1. Analyticity of f

The function $f(z)$ must be analytic at every point *inside* and *on* the contour C . This means f must be differentiable throughout the closed region bounded by C . If f has any singularities (poles, essential singularities, branch points) inside C , the standard formula does not apply directly – one must instead use the Residue Theorem.



2. Interior Point a

The point a at which we evaluate f must lie *strictly inside* the contour C . If a lies on C , the integral becomes improper and requires principal value interpretation. If a lies outside C , the integrand is analytic everywhere inside C , and by Cauchy-Goursat, the integral equals **zero**.



3. Positive Orientation

The contour C must be **positively oriented** – that is, traversed in the counterclockwise direction, so that the interior region lies to the left as one moves along C . If C is traversed clockwise (negative orientation), the result acquires a minus sign.

Geometric and Physical Interpretation

The Boundary Determines the Interior

The most profound insight of the Cauchy Integral Formula is this: **an analytic function has no "freedom" in its interior**. Once you know the values of $f(z)$ everywhere on a closed boundary curve C , the value of f at every point inside C is *uniquely and automatically determined*.

This is sometimes called the **rigidity property** of analytic functions. In physical terms, if you think of f as describing a steady-state temperature distribution or an electrostatic potential (both governed by Laplace's equation), the interior field is completely fixed by the boundary conditions – there is no additional information contained in the interior.

This property has no analogue for arbitrary smooth real-valued functions. A real function can be modified at a single interior point without affecting its boundary values. An analytic function cannot.

Mean Value Property

A particularly elegant special case arises when C is a circle $|z - a| = r$. Parameterizing $z = a + re^{i\theta}$ and substituting into the formula yields the

Gauss Mean Value Property:

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta$$

This states that $f(a)$ equals the *average* of f over any circle centered at a (provided the disk lies in the domain of analyticity). This is a powerful tool for deriving maximum modulus principles and Liouville's Theorem.

Maximum Modulus Principle: If f is analytic and non-constant in a domain, then $|f(z)|$ cannot attain a local maximum in the interior – the maximum must occur on the boundary.

Worked Example

Let us apply the Cauchy Integral Formula to a concrete problem. This example illustrates the mechanics of identifying the function $f(z)$, the point a , and the contour C , then applying the formula directly.

Problem Statement

Evaluate the contour integral:

$$\oint_C \frac{e^{3z}}{z - \pi i} dz$$

where C is the circle $|z| = 4$, traversed once in the positive (counterclockwise) direction.

Solution: Here $f(z) = e^{3z}$ is analytic inside and on the given circle $|z|=4$.

Also $z = a = \pi i$

$$\Rightarrow |z| = |\pi i| = \pi = 3.14 < 4$$

$\therefore z = \pi i$ lies inside the given circle .

Hence by Cauchy's integral formula we have,

$$\begin{aligned} \oint_C \frac{e^{3z}}{z - \pi i} dz &= 2\pi i f(\pi i) \\ &= 2\pi i e^{3\pi i} \\ &= 2\pi i (\cos 3\pi + i \sin 3\pi) \\ &= 2\pi i (-1 + 0) \\ &= -2\pi i. \end{aligned}$$

Question-2: Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z|=3$.

Solution:

$$\text{Let } \frac{1}{(z-1)(z-2)} = \frac{A}{z-2} + \frac{B}{z-1} \dots \dots \dots (1)$$

$$\Rightarrow 1 = A(z-1) + B(z-2) \dots \dots \dots (2)$$

Putting $z = 2$ and $z = 1$ in (2) we get $A = 1$ and $B = -1$

$$\text{Therefore } \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$$

$$\begin{aligned} \text{Hence } \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz \\ = \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} dz \\ - \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)} dz \dots \dots \dots (3) \end{aligned}$$

Let $f(z) = \sin \pi z^2 + \cos \pi z^2$. Since $z = 1$ and $z = 2$ are inside C and $f(z)$ is analytic inside and on C, so we can apply Cauchy's integral formula.

Now,

$$\begin{aligned} \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)} dz \\ = 2\pi i \{ \sin \pi (1)^2 + \cos \pi (1)^2 \} \\ = 2\pi i (\sin \pi + \cos \pi) \\ = -2\pi i \end{aligned}$$

$$\begin{aligned} \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} dz \\ = 2\pi i \{ \sin \pi (2)^2 + \cos \pi (2)^2 \} \\ = 2\pi i \end{aligned}$$

Substituting those values in (2), we get

$$\begin{aligned} \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} dz &= 2\pi i + 2\pi i \\ &= 4\pi i. \end{aligned}$$

Applications of the Cauchy Integral Formula

The Cauchy Integral Formula is far more than a theoretical curiosity — it is a practical workhorse with applications spanning pure mathematics, engineering analysis, and physics.



Evaluating Contour Integrals

The most direct application is computing complex contour integrals that would otherwise be intractable. By identifying the integrand, the integral reduces to evaluating a derivative at a point. This technique extends via the Residue Theorem to integrals of rational and trigonometric functions over the real line, a staple of engineering mathematics.



Harmonic Functions and Potential Theory

If $f = u + iv$ is analytic, both u and v are *harmonic functions* satisfying Laplace's equation. The Cauchy formula implies that harmonic functions also satisfy a mean value property. This is central to solving boundary value problems in electrostatics, fluid flow, and heat conduction — all areas of direct relevance to engineering.



Taylor and Laurent Series

The derivative formula provides the theoretical foundation for Taylor series expansions of analytic functions. By expressing as a contour integral, one proves that every analytic function equals its Taylor series within the disk of convergence. The Laurent series (for functions with isolated singularities) follows similarly, enabling series representations in annular regions.



Signal Processing and Control Theory

In electrical engineering, the Laplace and Fourier transforms are deeply connected to complex contour integration. The Cauchy formula underpins the inversion formulas for these transforms, enabling the analysis of linear time-invariant systems, filter design, and stability analysis via pole-zero plots in the complex plane.

Practice Problem

1. Show that $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, where C is the circle $|z|=5$.

2. Evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$, where C is the circle $|z|=4$.

3. Show that $\oint_C \frac{\sin \pi z + \cos \pi z}{(z-1)(z-2)} dz = 4\pi i$, where C is the circle $|z|=3$.